

Food Printing with Electrolysis Bubbles for Texture Control

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ABSTRACT

Research has been conducted regarding 3D food printing that additively manufactures edible objects. Some studies control the texture of printed foods by creating an intricate internal structure, which is limited by the resolution that can be printed with soft paste-like foods. This study proposes a method for controlling the texture of 3D-printed gummy candy by selectively generating fine foam structures inside the gummy using electrolysis. We establish a printing technique for generating bubbles in the gummy dough by determining the optimal material properties, developing an energization control system for electrolysis and implementing a design tool for the printing paths. Our results show that the ratio between the number of foamed layers and the total number of layers could control the ease with which breaking occurs while maintaining the original gummy's elasticity. We discuss the possibilities of exploring human-food interactions through our approach while demonstrating a gummy that contains foam structures in distinctive patterns.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Food printing, Food texture, Electrolysis, Foam, Bubble, Gummy

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1 INTRODUCTION

Food is not only a source of nutrition but also an important part of culture, and the exploration of new tastes, textures, and appearances has a long history. Food components are distributed across multiple phases, such as solids, liquids, and gases. Among them, gases (foam) have been used in various cuisines because they allow unique and rich food textures to be created. For example, sponge cakes and marshmallows have a fluffy softness, and the foam floating on the surface of a cappuccino has a unique mellowness. Another technique often used in French cuisine is that of foaming the sauce

to lend it a lighter texture and mouthfeel. In the field of human-computer interaction (HCI), there has been vigorous research into creating food that can be used as an interaction medium, including 3D food printing that additively manufactures edible objects with 3D printers. Recent technologies have made it possible for 3D printing to produce food with intricate shapes that are visually pleasing [22, 28]. There are also studies where the texture of 3D printed foods was controlled by modulating their intricate internal structure [18, 19, 21]. However, there is still room for further study of food texture using extrusion-based 3D food printing. In food printing, the printing resolution tends to be low, making it difficult to create fine internal structures. This is more pronounced when the printed food is soft, making it difficult for its structure to be maintained. Meanwhile, we believe that making a fine foam-like pattern inside an edible item would offer the advantage of creating a richer texture in the context of food printing.

To address this challenge, this research aims to make food texture controllable by selectively generating fine foam bubbles inside the food extruded during 3D food printing. Selective creation of food that includes foam is achieved through additive manufacturing while electrolyzing paste-like food material. Computer-controlled electrolysis can generate bubbles inside the food dough at arbitrary moments during the extrusion process. Additive manufacturing allows for the creation of foods in which the foam can be fixed in a designated position inside the material, because the food dough solidifies and can be layered with bubbles generated inside the dough. In this paper, as a first step, we propose a method for controlling the texture of gummy candy by creating bubbles in a gummy dough, which is a sol-like material suitable for electrolysis, while printing. To establish the printing technique for simultaneously printing and generating electrolysis bubbles in the gummy, the material properties of the gummy in respect of layering and electrolysis bubble generation are determined, and a special nozzle equipped with electrodes and an energization control system are developed. The developed software tool enables the generation of print files for realizing electrolysis at arbitrary moments during the printing to reflect a user's design of the foam area. We investigated the behavior of a gummy containing foam when it is pressed and found that the foam structure increased the gummy's softness and that the ratio between the number of foamed layers and the total number of layers could control the ease with which breaking occurs while maintaining the original elasticity of the gummy. We demonstrate that our method allows users to design and create gummies with a predetermined level of breakability at desired points in the material through several examples and discuss the contribution this method makes to HCI and human-food interactions (HFI) [2, 4], including its potential for allowing the design of dynamic textural experiences and special foods for people with chewing difficulties.

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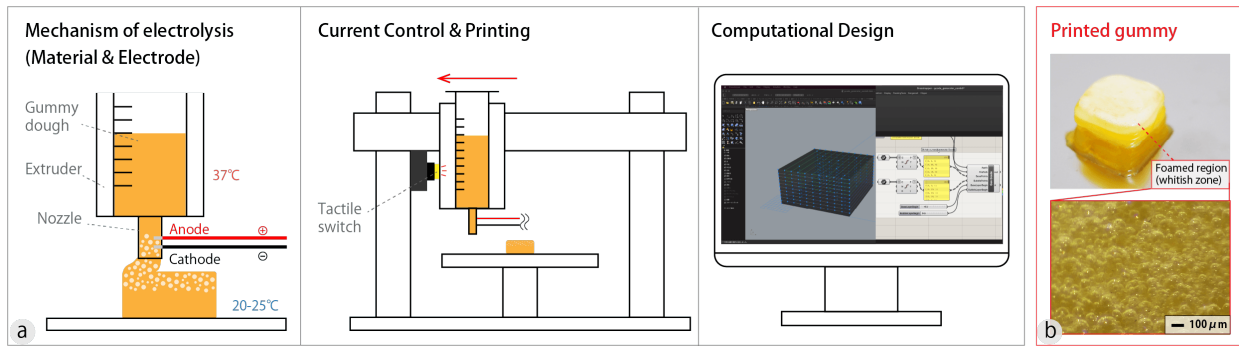


Figure 1: (a) Overview of the proposed method. (b) Gummy containing selective foam and foam architecture.

2 RELATED WORK

2.1 Aerated Food Gels

Bubbles in food have promising applications and have been utilized in the field of food science. In particular, nano-scale bubbles are used for food seasoning and improving the food freezing process [26]. In a fashion similar to our method, dispersed bubbles in edible gels - so-called “aerated gels” (AGs) - have potential applications in diet and gastronomic pursuits, and are the subject of ongoing research [30]. Methods for generating bubbles inside gels include those that involve dispersing gases with membranes, microdevices, ultrasound, and electrochemicals. Our method also involves an electrochemical process. In the field of HCI, research has been conducted to realize soft and deformable sensors and displays by mixing foam into bio-degradable gels [15, 16]. In previous implementations of AGs, the entire already existing food material was made into foam. In contrast, this study aims to create foods with distinctive textures by selectively including foam inside foods during food printing according to a user’s designed pattern.

2.2 3D Food Printing

Various approaches have been adopted in attempts to integrate foods with computers. These approaches include CNC milling to subtractively carve solid food into desired shapes [24], casting using pre-manufactured molds [5, 29], and stamping with digitally controlled jigs [17]. In the HCI field, extrusion-based additive manufacturing has been frequently utilized to print viscous and paste-like foods, including dough [9, 18, 23], tofu [24], and chocolate [14]. Also, 3D food printers are increasingly available commercially to help users create foods with the shapes and structures they want [3, 20]. Recent studies have aimed at printing a wider variety of shapes and applied the embedded printing method, which uses hydrogel baths as temporary scaffolds, to food printing [22, 28]. There are also studies that control the texture or mouthfeel of printed foods by modulating their infill construction [18, 19, 21]. While it has been reported that the texture could be controlled well enough to modify the chewing time [18], the internal structure is limited by the printing resolution that can be achieved with soft paste-like foods. The proposed method selectively performs electrolysis on the extruded food during the food printing process, which enables the construction of fine foam structures at arbitrary points inside the food. This allows users to design partial foam structures for use

in printing foods made of highly fluid dough, and to explore the possibility of realizing distinctive textures with a desired level of breakability.

This study uses gummy dough suitable for electrolysis in a combined printing and foaming process. There have been previous studies that investigated 3D printing methods for creating gummy materials and hydrogels using additive manufacturing techniques [6, 8], and this study also employs such a method. This is a technique in which a sol-like dough is extruded from a heated extruder, and cooled on a build plate to gelatinize, and then layered. This study is the first attempt to create a gummy with a foam structure by generating electrolysis bubbles while printing and layering.

2.3 Electrolysis in HCI

In the field of HCI, bubbles have long been used as pixels for information display. Conventionally, bubbles have been produced by supplying gas externally [7, 10] or by using gas contained in carbonated water [25], but electrolysis can alternate between foaming and non-foaming modes when the electric current powering it is switched on and off, making it possible to simply and controllably form a foamed structure through a computer-controlled process. Utilizing such advantages, bubble displays and color-forming displays embedded in small everyday objects such as coffee cups and decor [12, 13], and a high-pixel bubble display for art installation [11] have been proposed.

There is existing research that takes a similar approach to that adopted in this study, which involves printing objects while acting on the edible material at the nozzle by electrolysis [27]. This previous work focused on color control for 3D printing. The possibility of multi-color 3D printing for food has been explored by mixing edible pigments that change color according to pH (e.g., anthocyanins) with edible extruded material and selectively changing the pH of the material at the nozzle by electrolysis. In this study, we focus on the bubbles generated by electrolysis and aim to control the texture of foods by selectively creating foam structures inside the food.

3 SYSTEM OVERVIEW

The proposed method aims at creating rich textures by food printing, and exposes the extruded food to electrolysis during printing at arbitrary moments to selectively enclose foam inside the printed object (Figure 1). Additive manufacturing can solidify the food

dough and layer it as soon as the electrolysis bubbles are generated, enabling the foam to be fixed in a designated position¹. We use a commercially available 3D food printer and use gummy dough suitable for electrolysis as an extruded material. Electrolysis is performed at the nozzle through embedded electrodes. When the electrodes are energized during extrusion, electrolysis of water occurs, generating hydrogen and oxygen bubbles. Dough containing these bubbles is then layered, and a gummy that includes a foam structure can be created. By controlling the current flow appropriately, the area containing foam can be localized. The energization switch is triggered when the side of the extruder presses a tactile switch positioned on one side of the printer. The code for such special extruder movements is automatically generated via an implemented design tool that enables the area containing foam to be specified.

4 METHODS

4.1 Material and Electrode Preparation

A gummy dough was used as the extruded material because it is suitable for generating electrolysis bubbles. The gummy dough used in this study was made of 100% orange juice, gelatin, and granulated sugar. Sodium sulfate, a food additive, was added to the dough to promote current flow for electrolysis. The ratio of these ingredients by weight was 5:1:1:0.2 (juice:gelatin:granulated sugar:sodium sulfate). During the preparation of this material, the juice was heated in a microwave oven to over 60°C, and then each ingredient was added and mixed well. The dough prepared in this manner was confirmed to be capable of generating electrolysis bubbles. We employed the layering method reported in previous studies [6, 8], in which a dough extruded from a heated extruder is cooled at room temperature to gelatinize and then layered. The dough used is gelatin-based, so it is a flowing sol at high temperatures and a gel at temperatures below 25°C. By setting the extruder temperature to 37°C and the room temperature to 20–25°C, it was confirmed that the dough could be layered with gelation.

Electrolysis occurs at the nozzle to generate bubbles inside the gummy dough during extrusion. For this purpose, a 3D printed nozzle with an anode and a cathode for electrolysis was implemented (Figure 2). The nozzle was printed using ABS, and the inner diameter of the outlet was 1.6 mm. Two holes were drilled at 4 mm intervals on the side of the nozzle near the outlet, and an anode and a cathode were inserted into the nozzle through each hole. The holes were then filled with resin from the outside to fix the electrodes in place. Platinum wire was used for the anode to prevent elution via electrolysis. The cathode does not elute during electrolysis, so tinned copper wire was used because of its ready availability. Thus, no metal ions are released from the electrodes during electrolysis, and that the foamed gummies would not contain anything harmful to humans. Wiring was extended from these electrodes and connected to a current control circuit.

4.2 Current Control and Printing

In order to selectively generate foam in the printed object, it is necessary to switch between starting and stopping energization

¹Manual electrolysis of food in a mold can easily cause bubbles to flow out of their designated positions.

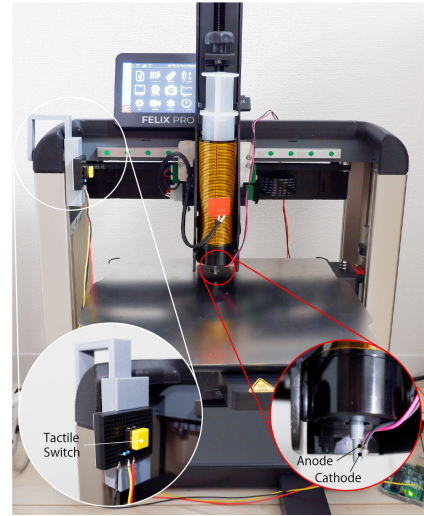


Figure 2: Printer setup. We implemented a special nozzle in which an anode and cathode were embedded. We placed the tactile switch system at the edge of the printer using 3D-printed fixation parts.

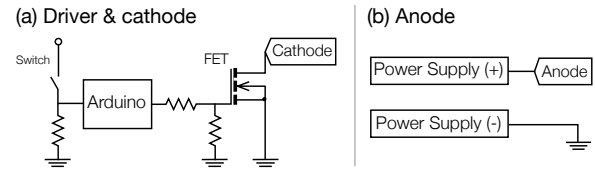


Figure 3: Circuit diagram for current control.

of the electrodes at arbitrary moments during printing. For this purpose, a tactile switch was installed at the edge of the printer (Figure 2), and a control circuit was implemented to switch the energization when the extruder moved and its side pressed against the switch (Figure 3). The cathode was connected to an Arduino UNO microcontroller board via a metal-oxide-semiconductor field-effect transistor (MOSFET)². When the extruder moved towards the tactile switch and pushed against it, the Arduino detected this and activated the MOSFET. When the MOSFET was activated, the cathode became connected to ground and hydrogen bubbles were generated from it. As shown in Figure 3 (b), the positive output of the DC power supply was constantly connected to the anode, and the negative output was connected to the common ground. Therefore, oxygen bubbles were also generated from the anode if the cathode was grounded and generating hydrogen bubbles. In this configuration, a gummy can be made as shown in Figure 1 (b) by applying a voltage of 27 V and a current of approximately 18 mA to the electrodes.

This study used a commercially available 3D food printer (FELIX SINGLE Food 3D Printer³). As described in the previous section, the printer's extruder was kept at 37 °C. The extruder was set to move at 3 mm/s to print a gummy containing sufficient foam. After each

²ON Semiconductor Corporation, 2N7000

³<https://www.felixfood.nl/felix-food-single-head>

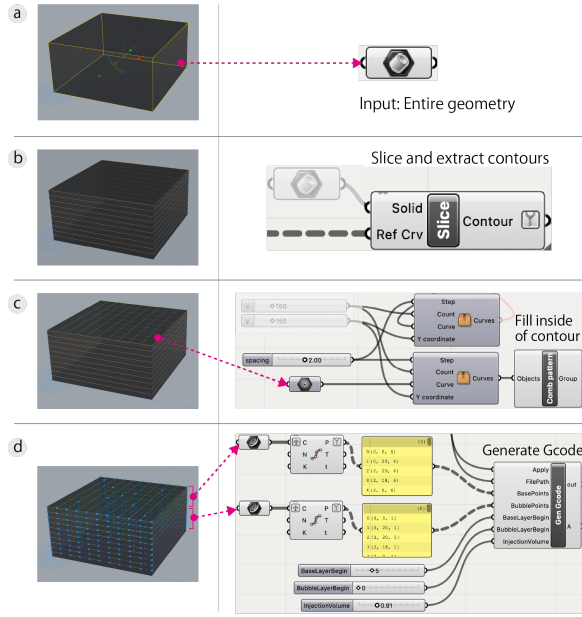


Figure 4: Design tool for the creation of printing paths. (a) The entire geometry of the printing object is created. (b) The tool automatically slices the specified geometry. (c) The tool automatically generates a comb pattern to fill the specified contour. (d) The tool automatically divides the specified comb curve at the vertices and generates printing paths. Note that the comb curves in green indicate the non-foamed regions and those in blue indicate the foamed regions.

layer was printed, printing was paused for 60 seconds to allow for sufficient gelation, avoiding the dough drips that prevent stacking in the height direction. Extra scanning and extrusion were performed after the tactile switch was pressed to eject dough in the nozzle after electrolysis was started or to eject foam in the nozzle after electrolysis was stopped. The printed gummies were stored in a refrigerator to gel thoroughly.

4.3 Computational Design Tool

A software system was implemented using Rhinoceros and Grasshopper. The system facilitates the design of objects for printing, including the specification of foaming region(s), and the export of finished designs as Gcode (Figure 4). The design process has three steps: (1) creating the entire geometry, (2) generating the printing paths, and (3) exporting to Gcode. First, the entire geometry of the object to be printed is designed. Then, when the created geometry has been specified in GrassHopper, that geometry is sliced layer by layer by extracting its contours. When a sliced contour is specified, a comb pattern is automatically generated to fill the inside of the closed contour region. This comb pattern is then specified for each foamed and non-foamed region. This divides the comb curve at the vertices and generates printing paths for each foamed and non-foamed region. Finally, by giving the position of the foaming layer as a parameter, the system generates Gcode commands by splicing the printing paths of the foamed and non-foamed regions. So that

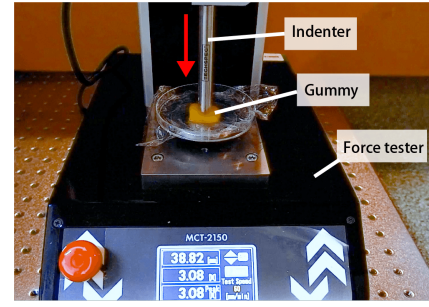


Figure 5: Experimental setup for force-displacement tests. The shape of this force tester’s indenter was a column, and the shape of the bottom of the indenter was circular with a diameter of 13 mm. The indenter descent speed was 1 mm/s.

the tactile switch for energization control described in the previous section would be activated by the side of the extruder, code was added to move the extruder to the switch activation position before and after the start and end of printing in the foamed region. Code was also added to run an extra scan after the switch was pressed and to pause for 60 seconds after each layer was printed.

5 EVALUATION

In order to evaluate the possibility of creating distinctive textures when the gummy containing selective foam was chewed, we investigated the behavior of the printed gummy when it was pressed. We conducted force-displacement tests for each ratio of the number of foaming layers to the total number of layers using a force tester⁴ (Figure 5). The conditions for the percentage of layers containing foam to be investigated were 30%, 50%, and 70%⁵; results for 0% (non-foamed) and 100% (completely foamed) were also compared. For each condition, gummies with foam in the top, middle, and bottom layers were created and tested (Figure 6 (a)). That is, in total there were 3 (percentage of foaming layers) $\times$ 3 (position of foaming layers) + 2 (non-foamed and completely foamed condition) = 11 conditions. We created three samples (rectangular gummy of W20 $\times$ D20 $\times$ H10 mm) for each condition. During this process, the current applied to the electrodes was approximately 18 mA, and the voltage was 27 V. Tests were conducted once per gummy, so three measurements were taken for each condition. Measurements were started when the surface temperature of each sample reached 15 °C.

The results showed that the ease with which the gummies would break varied with the proportion of layers containing foam, while the sense of repulsion due to elasticity of the gummies was found to be almost constant. Figure 6 (b) shows examples of force-displacement curves for each foaming ratio in the bottom layers. For all the conditions, the curves first show a gradual increase in force by elastic compression, followed by a gradual decrease in force by plastic deformation as the gummy breaks at the yield strength. After the force briefly decreases (after the repulsion point), it then increases again due to the gummy’s elasticity. The rough form of

⁴MCT-2150, A&D Co., Ltd.

⁵Examined gummies had foaming layers in three, five, and seven layers out of ten.

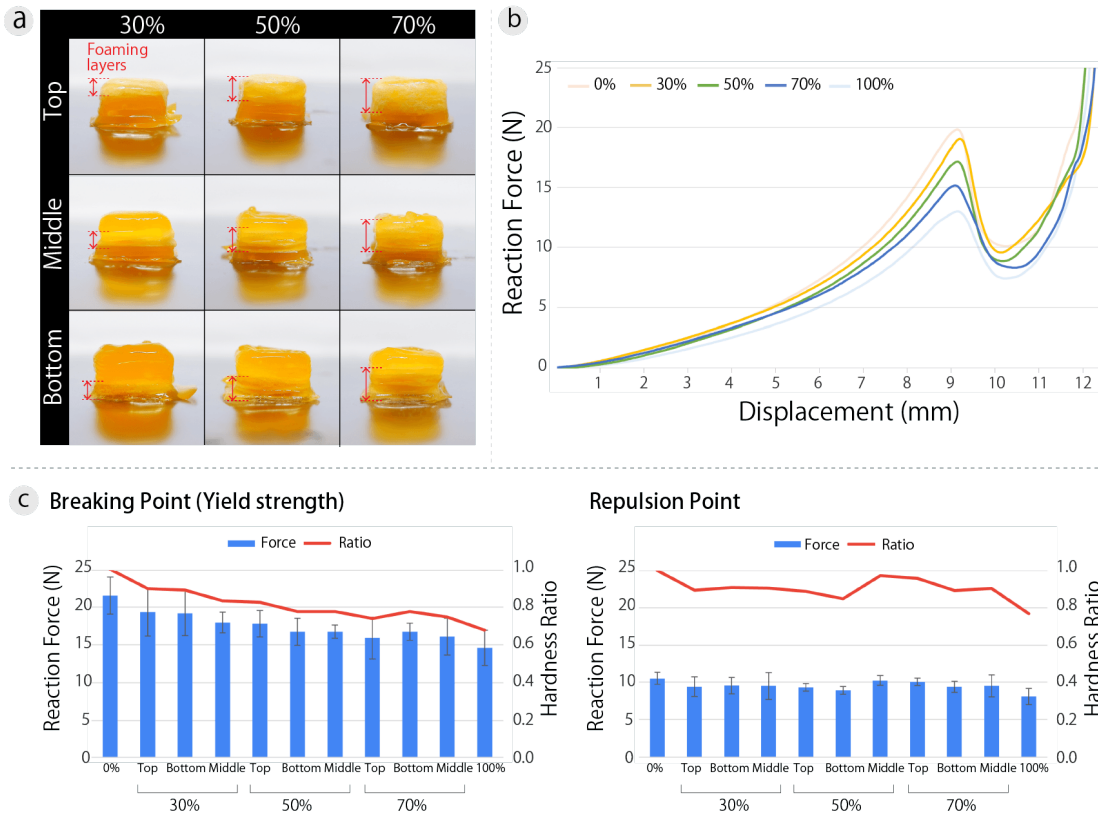


Figure 6: (a) Gummies were printed for each number of foaming layers and their location. Each column indicates the ratio between the number of foaming layers and the total number of layers, and each row indicates the position of the foaming layer. (b) Force-displacement curves for each ratio in the bottom layers. (c) Comparison of the averages of breaking points (yield strengths) and repulsion points for each condition.

this curve did not change depending on the gummy foam-layering ratio condition, but there were differences in the breaking point (yield strength) and the repulsion point of the curve for different foam-layering ratio conditions (Figure 6 (c)). Overall, the more layers containing foam, the smaller the breaking point, and the gummy becomes softer and easier to break. On the other hand, there was not much difference caused by the position of the layers containing foam, although there was some variation. This suggests that the layer(s) with foam can be placed at any position when producing a gummy of similar softness. The repulsion point for gummies under all conditions (except the 0% one) differed by less than 1 N from that of the gummy without foam (0% foam layer), regardless of the number and position of the foam layers, although the value was somewhat smaller in the case of the 100% foam layer. This can also be found in the force-displacement curves that indicate only a slight difference in repulsion points. We hypothesized that this was due to the presence of gummy layers without foam that exhibited high elasticity, even when they were very thin. We concluded that the foam structure increased the gummy's softness, and that the ease of breaking could be controlled by adjusting the ratio between the number of foamed layers and the total number of layers while maintaining the original gummy's elasticity to some extent.

6 DISCUSSION AND FUTURE WORK

Based on the results described in the previous section, by appropriately designing the foaming region, it is possible to fabricate gummies with a desired level of breakability and to realize food that provides novel experiences for HCI or HFI. For example, for people with chewing difficulties, such as infants, the elderly, and patients with temporo-mandibular joint disorders, our method enables the personalized fabrication of food products that are easy to break up according to the chewing ability of each person. In fact, the use of small gelatinous beads and foams is reported to inspire the creation of attractive texture-modified foods for the elderly [1], suggesting the applicability of our method to such a context. We believe that our method can also be used for training masticatory ability by providing foods that are easy to break up in the initial stage, and then gradually making them harder to fragment. Moreover, we can create novel eating experiences in the context of molecular gastronomy. For example, our method can create gummies that vary in hardness depending on the position and direction of the chewing motion by printing them with multiple layers each with different foam inclusion rates (Figure 7 (a)). Gummies that have a texture that corresponds to a visual appearance can also be created. Figure 7 (b) shows a turtle-shaped gummy containing foam only in

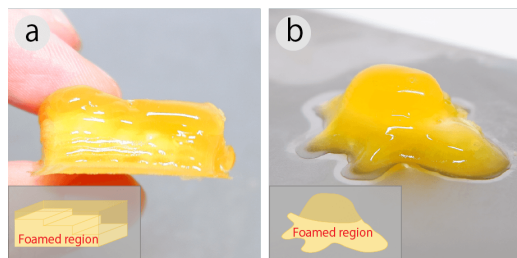


Figure 7: (a) Gummy having different degrees of softness depending on the chewing position. From left to right, the bottom 70%, 50%, and 30% layers contain foam. (b) Turtle-shaped gummy, containing foam only in the soft body parts.

the area of the body, while the area of the tortoiseshell was made of harder gummy without foam.

Future work includes further exploring the possibilities of our method for HFI by creating gummies with higher resolution and more complex foam patterns and investigating the user experience with such gummies. We believe that high-resolution foam printing can be achieved by placing electrodes near the nozzle outlet to prevent excess foam from accumulating in the nozzle and by introducing cooling systems such as cooling beds and fans to speed up the gelatinization of the foamed gummies. We also believe that our method can be applied to hydrogels other than gelatin, such as agar and carrageenan, and will explore the possibility of using these materials in technologies that are not limited to edible applications.

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